

GAUSE: Competition-Driven Specialization and Reward-Independent Capacity Assignment for Learner Populations

An Explanatory Companion: Architecture, Mechanisms, Demonstrations, and Applications

Yuhao Li*
University of Pennsylvania

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Abstract

Learning systems that must serve many distinct environmental regimes with capacity-limited learners face a persistent allocation problem: which learner handles which regime, and—when the environment is non-stationary—who *keeps* a regime that falls dormant? In practice this gap appears as three recurring failure modes: homogenization (every learner converges to the same behavior), coverage gaps (no learner masters rare regimes), and catastrophic forgetting (dormant-regime knowledge is overwritten to serve active ones). To address these failures, we present **GAUSE** (Generalist-averse Affinity Update for Specialist Emergence), a deliberately minimal population-learning framework built on three tightly coupled components: *per-regime belief learning* (Thompson sampling over a shared method inventory), *niche-affinity tracking* (a canonical exponentiated-gradient update on the regime simplex), and *winner-take-all competitive exclusion* (only the per-step winner reinforces its niche). At the core of the framework is a single emergent property with outsized consequences: the converged regime-to-agent assignment is *structural*—independent of current task reward—so a specialist simply idles while its niche is dormant and retains its expertise. We show that this property, not specialization per se, is what separates mechanisms that survive non-stationarity from mechanisms that do not: across five capacity-allocation arms (capacity-bounded monolith, learned Mixture-of-Experts router, fixed random niches, an EOI/CDS-style intrinsic diversity objective, and emergent competition), every reward-chasing allocation forgets dormant regimes and every reward-independent assignment retains them (+71% lower post-reactivation error, $p \sim 10^{-35}$; robust to swapping hard eviction for soft interference). Empirical results across six real-world domains (145,294 records) and controlled synthetic tasks show that the framework reliably self-organizes into one-per-regime specialists (mean SI = 0.65 from competition alone), matches purpose-built diversity and routing baselines on spatial coverage with none of their machinery, and is structurally immune to a failure mode of reward-driven routing. We close with a survey of application domains—elastic memory for agentic AI, regime-switching finance, edge sensing, robot fleets, and serving infrastructure—where the mechanism’s parsimony and retention guarantee are most valuable. Code and data: <https://github.com/HowardLiYH/GAUSE>.

Keywords: emergent specialization; competitive exclusion; multi-agent learning; catastrophic forgetting; mixture-of-experts; reward-independent assignment

*Contact: Yuhao Li (li88@sas.upenn.edu).

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1 Introduction

“Two species with identical ecological niches cannot stably coexist.”
—the competitive exclusion principle [5, 8]

Populations of learners that share an environment—trading strategies in a market, sensors in a distributed network, experts inside a mixture-of-experts model—face a coordination problem that single learners do not: how should the population *divide* a heterogeneous task space, without a central coordinator, so that every regime of the environment is served by a competent specialist? Despite decades of progress in multi-agent learning [27], three intertwined limitations persist in standard approaches.

First, **shared objectives homogenize**. Cooperative multi-agent reinforcement learning methods (IQL [23], VDN, QMIX [22], MAPPO [26]) optimize a shared task return, and a well-documented consequence is that agents converge to near-identical policies. In our head-to-head re-runs, all four methods reach a Specialization Index below 0.02 on every domain—no division of labor emerges by default. Second, **explicit coordination is expensive**. Methods that do produce diverse roles—diversity-MARL (ROMA [24], RODE [25], MAVEN [18], EOI [9], CDS [14]), quality-diversity archives [19, 21], or learned gating networks—require auxiliary machinery: intrinsic identifiability rewards, behavior descriptors chosen *a priori*, or a trained router. Each adds objectives, hyperparameters, and failure modes of its own. Third, and least appreciated, **reward-driven capacity allocation forgets** [3, 11]. When the environment is non-stationary and per-learner capacity is bounded, any mechanism that allocates capacity by chasing current reward—a single capacity-bounded model, or a Mixture-of-Experts gate trained on task reward—systematically overwrites knowledge of regimes that are currently dormant, and pays a full relearning cost each time a regime reactivates. The dormant regime is precisely the one emitting no reward signal, so a reward-driven allocator receives no signal to protect it.

To address these limitations, we introduce a cohesive design philosophy centered on a single ecological mechanism: **competitive exclusion** [5, 8, 17]. The framework, named **GAUSE** (Generalist-averse Affinity Update for Specialist Emergence)¹, is built from three integrated components (Figure 1):

First, each agent maintains *per-regime method beliefs*—Beta distributions over the success of each prediction method in each regime, acted on by Thompson sampling. This is the agent’s competence: what it knows how to do, and where. **Second**, each agent tracks a *niche affinity* distribution over regimes, updated by the canonical exponentiated-gradient (Hedge / multiplicative-weights) rule [12, 4, 1]. This is the agent’s identity: where it belongs. **Third**, agents are coupled solely through *winner-take-all competition*: at each step, all agents act, and only the best performer reinforces its affinity for the current regime. No communication, no shared reward, no gate, no diversity objective. Competition makes homogeneous strategies unstable—identical agents split the same wins, so deviating to a less-contested regime pays—and the population partitions into one-per-regime specialists.

The key insight of this document is that the resulting assignment has a property that purpose-built alternatives lack: *it is reward-independent at convergence*. A converged specialist holds its niche by identity, not by moment-to-moment reward, so when its regime goes dormant the specialist simply idles and *retains* its expertise (Figure 4). We formalize this as a **reward-independence principle** (Section 7), show that it cleanly dichotomizes five capacity-allocation mechanisms in controlled experiments (every reward-chasing arm forgets; every reward-independent arm retains), and demonstrate that it independently corroborates recent mixture-of-experts theory [15], which finds that the gating network’s updates must be *terminated* for continual-learning convergence.

In summary, this work makes the following key contributions:

1) A reward-independence principle for retention. We identify reward-independence of capacity assignment as the property governing whether a capacity-bounded system retains dormant-task knowledge under non-stationarity, verify it five-for-five across allocation mechanisms, and explain it with an idealized observation: dormant tasks emit no protective reward signal, so only structural assignments can reserve capacity for them.

2) A coordination-free specialization mechanism. GAUSE couples winner-take-all competition with an exponentiated-gradient niche update and nothing else. Competition alone (niche bonus $\lambda = 0$)

¹The name also honors G. F. Gause, whose competitive exclusion principle (1934) the mechanism operationalizes.

drives mean SI to 0.65 across six real domains; with the default bonus the population reaches $SI \approx 0.99$ and one-per-regime coverage.

3) Honest benchmarking against purpose-built baselines. Rather than compare only to cooperative MARL (which does not try to specialize), we compare to an EOI/CDS-style learned-diversity objective and a learned MoE router at matched capacity. Spatial coverage proves to be a commodity all three achieve; competition’s value there is parsimony. Retention is not a commodity: the reward-driven router fails it decisively.

4) A meaningfulness audit. We show that the Specialization Index certifies *organization*, not exploitable structure (it is invariant to regime-label shuffling and inflatable by the learning rate), and we localize exactly when division of labor improves task accuracy—under bounded capacity and non-stationarity—and when it does not.

2 Related Work

We review prior work along five dimensions closely related to GAUSE.

Cooperative MARL. Independent Q-Learning [23], value-decomposition methods (VDN, QMIX [22]), and multi-agent policy gradients (MAPPO [26]) optimize shared objectives with centralized training. These methods are the right baselines for cooperative control, but they do not attempt diversity, and empirically they homogenize: in our re-runs all reach $SI \leq 0.02$. We treat them as confirmatory contrast, not as competitors on specialization.

Diversity- and role-emergence MARL. ROMA [24] and RODE [25] learn role embeddings; MAVEN [18] injects diverse exploration; EOI [9] and CDS [14] add intrinsic individuality or mutual-information objectives; R3DM [6] discovers roles through dynamics models. These methods do produce division of labor, at the cost of explicit diversity machinery. We benchmark directly against an EOI/CDS-style intrinsic-reward arm and find it a strong baseline: it matches competition on coverage for $K \geq 2$ and retains dormant regimes (its assignment is driven by an identity reward, not task reward—the reward-independence principle again). Competition’s advantage over it is parsimony, not dominance.

Quality-diversity optimization. MAP-Elites [19] and novelty search [13] maintain archives of diverse high performers, but require behavior descriptors defined *a priori*. GAUSE needs no descriptors: niches are discovered by the dynamics.

Mixture-of-experts in continual learning. Recent theory [15] establishes that MoE can mitigate catastrophic forgetting [3, 20] by isolating tasks in experts—*provided* experts are plentiful relative to tasks and the gating network’s updates are eventually terminated, without which continual task arrival destabilizes routing. Our experiments occupy exactly the regime those provisos exclude (fixed total capacity, experts forced into reuse, gate still learning) and show that isolation then breaks down: the router inherits the monolith’s forgetting. Freezing the gate is, in our terms, making the assignment reward-independent; competition reaches that state emergently, with no freezing schedule to tune.

Ecological niche theory. The competitive exclusion principle [5, 8] and MacArthur’s resource partitioning [17] describe how competition alone drives species into complementary niches [7]. GAUSE operationalizes this for artificial learners and inherits its most useful property: the partition, once formed, is maintained by identity rather than by ongoing payoff, which is what makes it robust to dormancy.

3 The GAUSE Framework: A Unified Architecture for Emergent Division of Labor

This section presents the integrated architecture. We first articulate the design philosophy (3.1), then detail the architectural blueprint (3.2), and finally explain the per-step dynamics that realize self-organization (3.3). Figure 1 contrasts the standard reward-driven allocation stack with the GAUSE architecture.

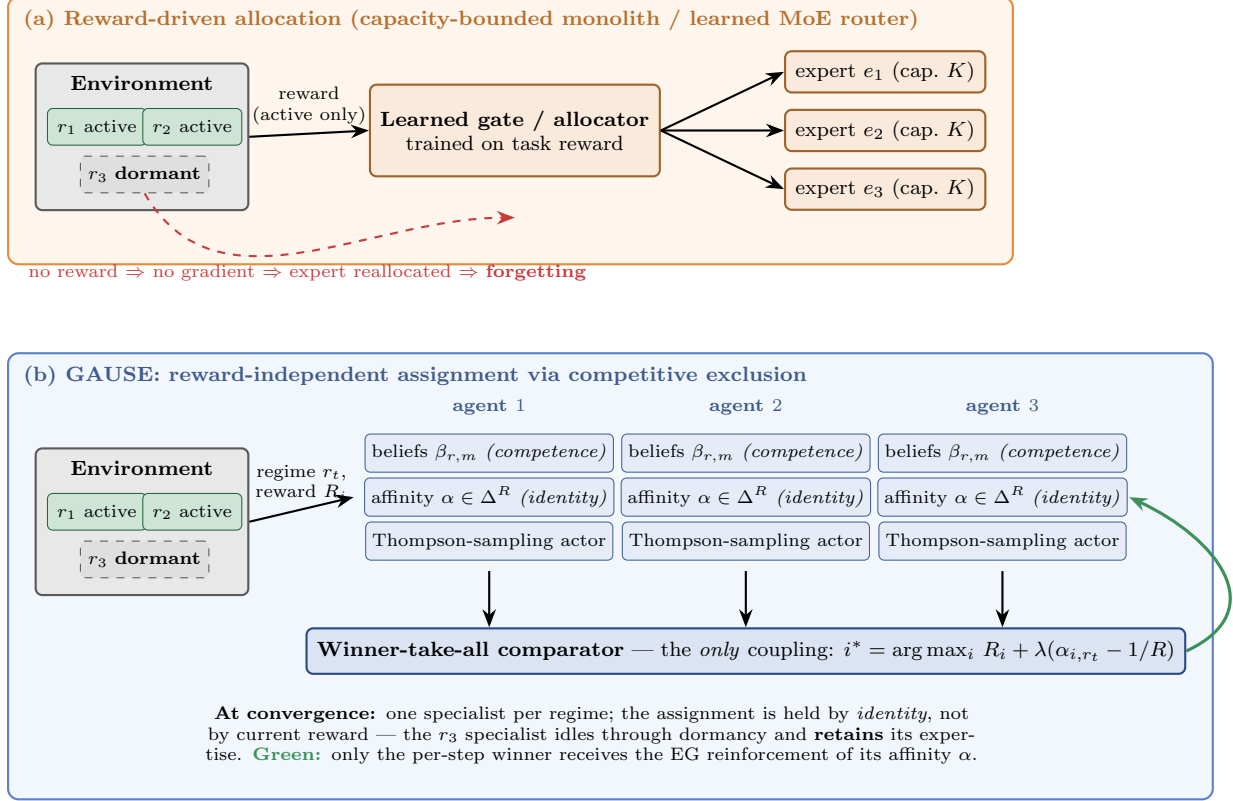


Figure 1 The GAUSE architecture, contrasted with reward-driven allocation. (a) A gate (or a single bounded model) assigns capacity by chasing current task reward; a dormant regime emits no reward gradient, so its capacity is silently reallocated and must be relearned on reactivation. (b) GAUSE couples agents only through per-step winner-take-all competition; each agent’s niche affinity α concentrates on the regime it wins, and the converged one-per-regime partition is *structural*—it persists with no reward flow, which is the source of the retention results in Section 8.

3.1 Design Philosophy: The Competitive-Exclusion Loop

Contemporary approaches to division of labor treat the assignment of learners to tasks as something to be *optimized*: a gate is trained, a diversity objective is added, roles are learned. GAUSE is founded on the opposite principle: **the assignment should be an equilibrium, not an objective**. We formalize this as the competitive-exclusion loop, a recursive paradigm structured around four functions (Figure 2):

- Compete**: all agents act on the current regime; one wins.
- Reinforce**: only the winner strengthens its affinity for that regime.
- Differentiate**: losers’ relative affinities drift toward less-contested regimes.
- Stabilize**: once each agent owns a regime, deviation is unprofitable; the partition is a fixed point.

The key property of this loop is *what it leaves behind*. Because the converged partition is a fixed point of identity (each agent’s affinity concentrates on its niche), it no longer depends on the flow of reward. An agent whose niche is dormant receives no wins, but it also receives no pressure to abandon its niche—it simply idles. This is the structural source of the retention results in Section 8: the loop intrinsically unifies *organizing* with *remembering*.

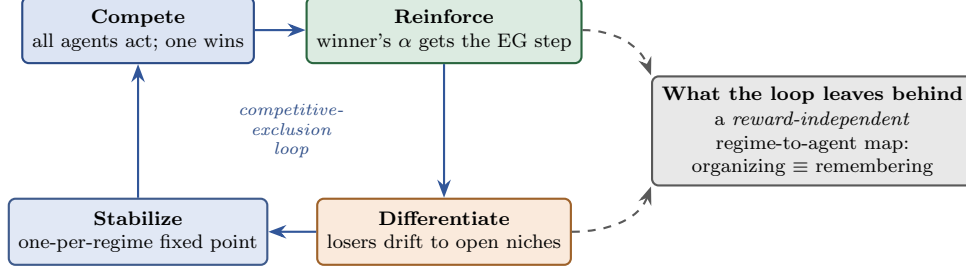


Figure 2 The competitive-exclusion loop. Compete–Reinforce–Differentiate iterates until Stabilize: each agent owns a regime and deviation is unprofitable. The loop’s by-product is its most valuable output—a converged assignment that no longer depends on the flow of reward.

3.2 Architectural Blueprint: Components and Their Roles

The loop is instantiated through three principal components per agent, plus one population-level rule (Figure 1b).

1. Method-Belief Store: The Competence Layer. Each agent i maintains, for each regime r it has encountered, a Beta distribution $\text{Beta}(\beta_{i,r,m}^+, \beta_{i,r,m}^-)$ over the success rate of each method m in a shared inventory of M methods. This store answers “*what works here?*” and is updated by standard Bayesian counting. In the bounded-capacity experiments the store is explicitly finite: it holds at most K regimes, with either least-recently-used eviction (hard model) or graded interference decay (soft model)—Section 6.

2. Niche Affinity: The Identity Layer. Each agent maintains a distribution $\alpha_i \in \Delta^R$ over regimes—its evolving answer to “*where do I belong?*” Affinity is updated only on wins, by the canonical exponentiated-gradient rule (Section 4), and read out through the Specialization Index $\text{SI}(\alpha) = 1 - H(\alpha)/\log R$, which is 1 for a one-hot specialist and 0 for a uniform generalist.

3. Thompson-Sampling Actor: The Decision Layer. At each step the agent samples $\tilde{\theta}_m \sim \text{Beta}(\beta^+, \beta^-)$ for every method and plays the argmax—a principled exploration–exploitation balance requiring no tuned exploration schedule.

4. Winner-Take-All Rule: The Population Coupling. The *only* interaction between agents is the per-step competition: the agent with the highest (optionally niche-shaped) reward is declared winner and is the only one to reinforce its affinity. There is no communication channel, no shared critic, no gate, and no diversity term. This minimality is deliberate: every emergent property of the system must be attributable to competition, because nothing else couples the agents.

3.3 System Dynamics: One Step of the Loop

The components interact in a fast per-step cycle. Given the environment in regime r_t :

1. **Act.** Every agent i selects a method m_i by Thompson sampling from its beliefs for r_t (agents without competence in r_t act uninformed) and receives raw reward R_i .
2. **Score.** Each agent’s competitive score adds an additive niche bonus centred on the uniform affinity: $\tilde{R}_i = R_i + \lambda (\alpha_{i,r_t} - 1/R)$, with $\lambda \geq 0$. At $\lambda = 0$ the bonus vanishes and raw reward decides.
3. **Select winner.** $i^* = \arg \max_i \tilde{R}_i$.
4. **Update beliefs.** Agents update their Beta counts for the methods they played (success/failure), within their capacity budget.
5. **Update identity.** Only the winner applies the exponentiated-gradient affinity update for r_t and renormalizes.

Two timescales emerge from this single cycle without being designed in: a *fast* competence dynamic (beliefs sharpen within a regime in tens of steps) and a *slow* identity dynamic (affinities concentrate over hundreds of steps). The slow dynamic is the one with the architectural consequence: once identities have

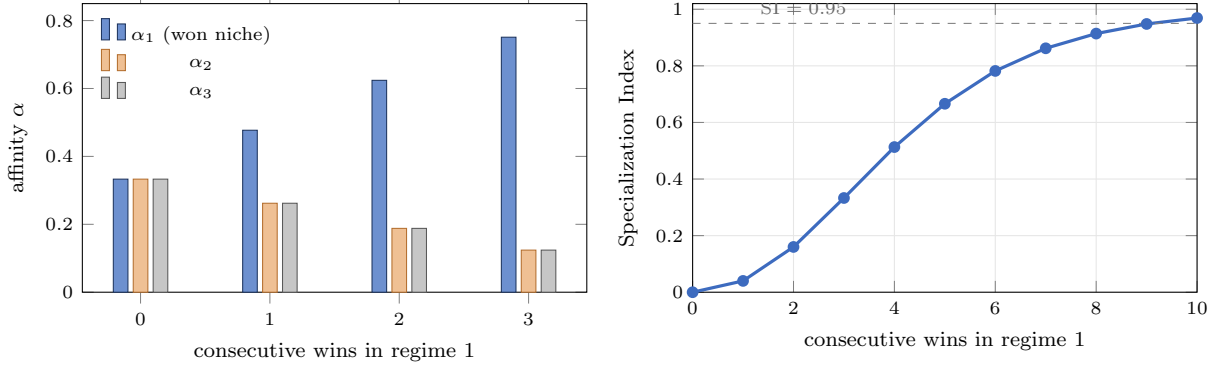


Figure 3 Worked demonstration of the exponentiated-gradient affinity update ($R = 3$, $\eta = 0.6$, quality $q = 1$). **Left:** the winning regime’s affinity grows multiplicatively while the others shrink by renormalization. **Right:** the Specialization Index $SI = 1 - H(\alpha)/\log R$ traced over ten consecutive wins—identity concentrates quickly, and only positive evidence (winning) ever moves it.

concentrated, the regime-to-agent map is fixed and reward-independent, and the population behaves as a distributed, persistent memory over the regime space.

4 Niche Affinity and the Exponentiated-Gradient Update

4.1 The Update Rule

When agent i wins in regime r with normalized reward (quality) $q \in [0, 1]$, its affinity updates multiplicatively and renormalizes:

$$\alpha_{i,r} \leftarrow \frac{\alpha_{i,r} e^{\eta q}}{\sum_{r'} \alpha_{i,r'} e^{\eta q} \mathbf{1}_{[r'=r]}}, \quad (1)$$

the canonical exponentiated-gradient / Hedge / multiplicative-weights step on the regime simplex [12, 4, 1, 2]. Three properties motivate this choice over additive heuristics. **(i) Simplex preservation by construction:** affinities remain a probability distribution with no clamping (an earlier additive variant of this project required an undocumented clamp whose pathology suppressed roughly half the specialization signal; replacing it with EG is what separates the current V4 results from the V3 ones). **(ii) A constant effective rate:** the multiplicative step does not slow down or speed up depending on the current state in the way the clamped additive rule did. **(iii) A standard form:** EG on the simplex is the textbook update for tracking the best coordinate, making the algorithm’s one learnable dynamic maximally legible. We deliberately do *not* claim Hedge’s adversarial regret bound transfers to our coupled winner-take-all setting; that would require separate analysis.

4.2 Worked Demonstration

Consider $R = 3$ regimes, $\eta = 0.6$, and an agent at uniform affinity $\alpha = (0.333, 0.333, 0.333)$. The agent wins three consecutive steps in regime 1 with quality $q = 1$:

$$(0.333, 0.333, 0.333) \rightarrow (0.477, 0.262, 0.262) \rightarrow (0.624, 0.188, 0.188) \rightarrow (0.751, 0.124, 0.124).$$

Three wins move SI from 0 to 0.33; ten wins take it past 0.95 (Figure 3). The same trajectory in reverse never occurs spontaneously: losing in regime 1 does not push the agent away (losers do not update); it is *winning elsewhere* that re-anchors identity. This asymmetry—identity moves only on positive evidence of fit—is what makes converged identities stable through dormancy.

4.3 Heterogeneous Initialization and Symmetry Breaking

Agents start near-uniform with small random perturbations. The perturbation carries no labels and no assignment; it only breaks ties so that the first few competitions are not decided by coin flips alone. The eventual one-per-regime partition is an equilibrium selected by the dynamics, not an initialization artifact: random-shuffle controls confirm the same partition statistics under re-drawn perturbations.

5 Competitive Exclusion and Winner-Take-All Dynamics

5.1 Why Strictness Matters

Winner-take-all is not a simplification of softer competition—it is the key structural driver of differentiation. Under proportional (softmax) credit, identical agents can persist indefinitely: each receives a share of every regime’s reward, and hedging across niches is stable. Under winner-take-all, identical agents split wins evenly in expectation, so each one’s per-regime reinforcement halves; an agent that deviates to an uncontested regime wins that regime *outright*. Formally (companion paper [16], Proposition 1), homogeneous strategy profiles are not Nash equilibria of the winner-take-all game: a unilateral deviation to a less-contested niche strictly increases expected reinforcement.

5.2 The Niche Bonus Is an Accelerator, Not the Cause

The additive shaping term $\lambda(\alpha_{i,r_t} - 1/R)$ advantages an agent competing in a regime it already prefers. Setting $\lambda = 0$ removes it entirely—and specialization still emerges, with mean SI 0.65 across the six real domains (every domain > 0.49 , versus 0.13 for a random baseline). The λ -sweep is flat and high (SI > 0.98) across $\lambda \in [0.2, 0.4]$ and declines at 0.5, where over-locked winners stop contesting secondary regimes and leave some niches under-filled. The mechanism therefore needs no shaping to work; shaping only sharpens it.

5.3 What the Population Looks Like After Convergence

On every real domain, the converged population is a clean partition: each agent holds a primary niche with affinity ≥ 0.95 , every regime is owned by at least one agent, and the population collectively uses 86% of the shared method inventory (specialists adopt the methods suited to their niche). The assignments are interpretable by inspection—a property no deep MARL baseline in our comparison offers.

6 Bounded Capacity and Memory Models

Specialization only matters when no single learner can do everything. We model this with an explicit per-agent capacity K : each agent can hold beliefs for at most K of the R regimes. Two memory models implement the constraint.

6.1 Hard Model: LRU Slots

The belief store holds K regime entries; learning an uncached regime evicts the least-recently-used entry. Eviction is total: the evicted regime’s beliefs reset to the uninformed prior, and re-encountering it costs a full relearning period. This is the cleanest possible model of “capacity binds.”

6.2 Soft Model: Interference Decay

A natural objection is that real learners do not evict discretely—knowledge *interferes* [3]. The soft model removes eviction entirely: every learning update on the active regime decays the pseudo-counts of the agent’s other held regimes by a factor that grows with how far the agent’s learned footprint exceeds K . Dormant regimes are never refreshed, so they decay smoothly toward the prior—the interference account of catastrophic forgetting, with graded rather than all-or-nothing loss. Every retention result in Section 8

is reproduced under both models; the headline separation is a property of *who gets updated*, not of how forgetting is discretized.

6.3 The Five Capacity-Allocation Arms

All experiments compare five mechanisms at matched per-agent capacity K :

1. **Monolith**: one agent with capacity K serves everything. Allocation chases the active regime by necessity.
2. **MoE router**: N capacity- K experts behind a learned gate (ϵ -greedy routing; gate trained on routed-expert quality). Allocation chases current reward by design.
3. **Random diversity**: N agents with fixed, random capacity- K niches. Allocation is frozen—fully reward-independent, but coverage is accidental.
4. **Learned diversity (EOI/CDS-style)**: all agents receive an intrinsic identifiability reward $\log p(i | r)$ on top of task quality [9, 14]; the identified owner learns the regime. Allocation follows an *identity* signal, not task reward.
5. **GAUSE (ours)**: winner-take-all competition; allocation is an emergent, converged identity.

This five-arm design is deliberate: the arms differ in *what signal drives capacity assignment*, which is exactly the axis the next section’s principle predicts should matter.

7 The Reward-Independence Principle

7.1 Why a Reward-Driven Router Forgets

Consider a regime r that goes dormant for an extended interval while an expert e holding r receives D learning updates from other, active regimes. Figure 4 illustrates the mechanism.

Observation 1 (Dormant regimes receive no protective signal). *Under the hard capacity model, e retains r only if fewer than K distinct regimes are used during the interval; under the soft model, r ’s retained strength decays as $(1 - \rho)^D \rightarrow 0$. Crucially, a dormant regime emits no steps and hence no reward gradient: the gate update $\Delta \text{gate}[r][e] \propto (q_e - \frac{1}{2})$ is identically zero throughout dormancy, so the router receives no signal that would lead it to reserve e for r . Reactivation therefore incurs the full relearning cost with probability $\rightarrow 1$ as dormancy grows.*

The asymmetry is the crux: *routing and capacity protection draw on the same signal, and the regime most needing protection is the one producing none.* A converged competitive (or identity-driven) assignment escapes the trap because it is reward-independent at steady state: the specialist for r simply idles through dormancy, applying zero foreign updates to its own store ($D = 0$ for its niche), and retains it structurally—no signal required.

Principle 1 (Reward-independent assignment). *Under bounded per-agent capacity and non-stationarity, a capacity-allocation mechanism retains dormant-task knowledge if and only if its regime-to-learner assignment is independent of current task reward.*

7.2 The Five-Arm Dissociation

Principle 1 makes a sharp prediction across the five arms of Section 6, verified in full by the experiments (Table 1):

Five for five, retention tracks reward-independence—not specialization, not population size, not learning sophistication. Among the retaining arms, competition is the most parsimonious: random diversity retains but cannot guarantee coverage at $K = 1$ (unowned regimes are possible); learned diversity retains but requires the auxiliary intrinsic-reward machinery; gate-freezing (the prescription of recent MoE continual-learning theory [15]) retains but requires deciding *when* to freeze. Competition needs none of these—the reward-independent assignment is the equilibrium the dynamics converge to.

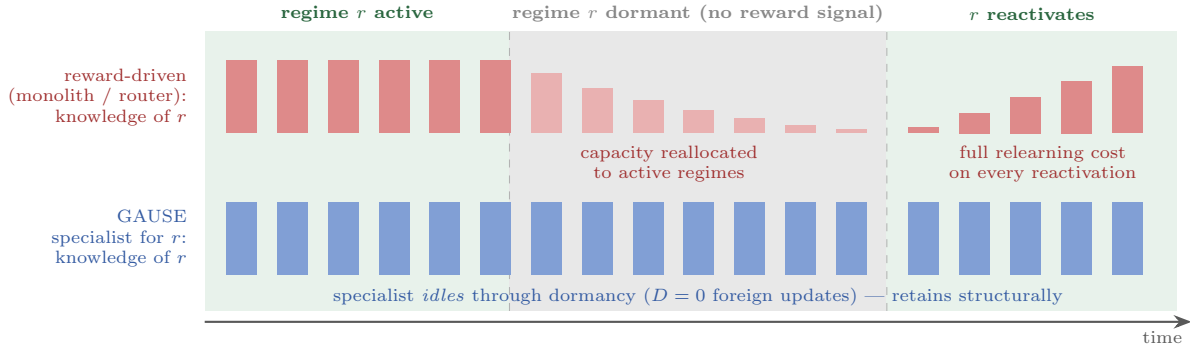


Figure 4 Why reward-independence governs retention. **Top:** a reward-driven allocator (capacity-bounded monolith or learned MoE router) reallocates the dormant regime’s capacity to active regimes—the dormant regime emits no reward gradient, so nothing protects it—and pays a full relearning cost at reactivation. **Bottom:** a converged GAUSE specialist holds its niche by identity, applies no foreign updates to its own store during dormancy, and is immediately competent on reactivation. Measured post-reactivation errors for all five arms appear in Table 1 and Figure 7.

Arm	Assignment signal	Reward-indep.?	Retains?
Monolith	current active regime	no	no
MoE router	current task reward	no	no
Random diversity	none (frozen)	yes	yes
Learned diversity	intrinsic identity reward	yes	yes
GAUSE	converged identity	yes	yes

Table 1 The five-arm dissociation predicted by Principle 1. Retention tracks reward-independence of the assignment signal five-for-five; it does not track specialization, population size, or learning sophistication.

7.3 Relation to MoE Continual-Learning Theory

Recent theory on mixture-of-experts in continual learning [15] proves that the gating network’s updates must be *terminated* after sufficient training for the system to converge, and that MoE’s forgetting-mitigation holds when experts are plentiful or can be added. Read through Principle 1, both findings are the same statement: a stabilized (frozen) gate is a reward-independent assignment, and plentiful experts remove the forced reuse that makes reward-chasing destructive. Our experiments cover the complementary regime the theory excludes—fixed total capacity, forced reuse ($K \ll R$), gate still learning—and find exactly the failure the principle predicts. The two lines of work corroborate each other from opposite directions: regression theory prescribes freezing; population dynamics exhibits a mechanism that arrives at the frozen state emergently.

8 Experiments and Evaluation

8.1 Experimental Settings

Domains. Six real-world domains with verified public data (145,294 records): cryptocurrency (Bybit; 8,766 daily bars), commodities (FRED; 5,630), weather (Open-Meteo; 9,105), solar irradiance (Open-Meteo; 116,834 hourly), urban traffic (NYC TLC; 2,879 hourly), and air quality (Open-Meteo; 2,880 hourly). Each domain defines 4–6 regimes and 5 tailored prediction methods. Controlled synthetic regime-champion streams (each regime has a distinct optimal method, with a tunable separation parameter) isolate mechanisms that real data confounds.

Baselines. Homogeneous (best single method), random selection, cooperative MARL (IQL, VDN, QMIX, MAPPO) at matched agent counts and training budgets, and—for the capacity experiments—the five-arm comparison of Section 6.

Metrics. Specialization Index (organization), task prediction error (performance), post-reactivation error (retention, measured in a 15-step probe window after each regime reactivation). Protocol: 20–30 seeded

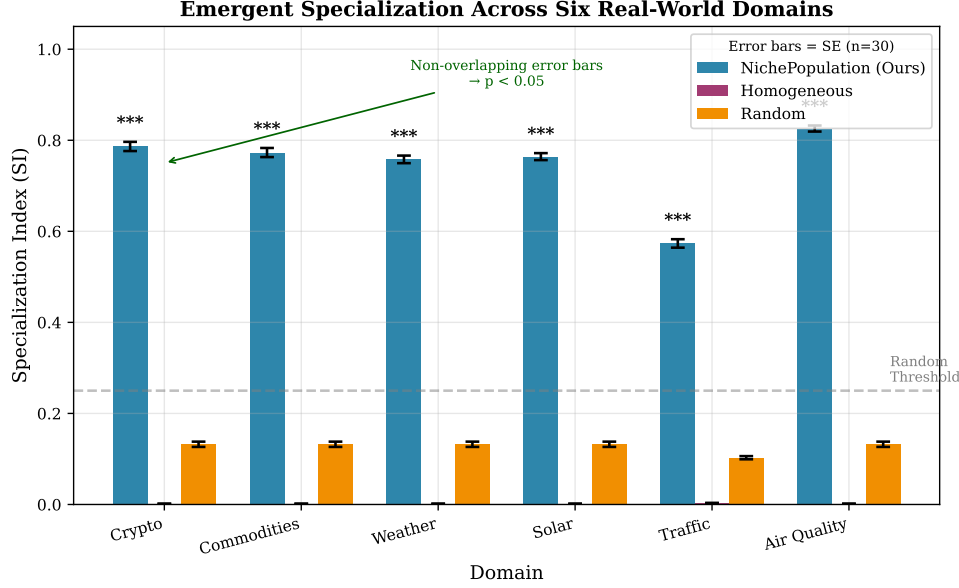


Figure 5 Emergent self-organization across the six real-world domains: GAUSE reaches $SI \approx 0.99$ everywhere, while homogeneous and random baselines stay near zero. We treat this organization result as a diagnostic, not a headline—the audit (Section 8.6) shows what SI does and does not certify.

trials per condition, 95% confidence intervals, one-sided Welch tests with Holm–Bonferroni robustness checks for sweep families.

8.2 Results on Emergent Self-Organization

Under the default configuration, every domain converges to near-perfect organization (mean $SI = 0.992$; all domains > 0.98 ; Figure 5), versus ≤ 0.02 for all cooperative MARL baselines—a qualitative, not quantitative, gap. The critical ablation sets $\lambda = 0$: competition alone yields mean $SI = 0.65$ (every domain > 0.49), $5.1\times$ the random baseline, confirming that the niche bonus accelerates but does not cause specialization. Populations use 86% of the method inventory on average; division of labor extends to *how* agents predict, not just *where*.

We treat these SI results as a diagnostic, not a headline. A battery of controls shows SI is invariant to regime-label shuffling and inflatable by the learning rate: high SI certifies organization, not exploitable structure. The performance-relevant results follow.

8.3 Results on Coverage Under Bounded Capacity

With per-agent capacity bounded ($K < R$), the specialized population beats the capacity-matched monolith decisively (synthetic $+73\%$ at $K=1$; real traffic $+22\%$), and beats the equal-capacity *random*-diversity population specifically at tight capacity (synthetic $+57\%$, $p < 10^{-16}$; traffic $+8.7\%$, $p < 10^{-3}$ at $K=1$), where random assignment leaves coverage gaps that competitive exclusion provably fills (Figure 6). Against the purpose-built arms the honest finding is parity: the learned-diversity objective matches competition for $K \geq 2$ (and exceeds it on some real domains), and the MoE router matches it on the synthetic task at every K . A method-overlap sweep maps the boundary: competition’s edge over learned diversity grows monotonically with method exclusivity, from -4.8% (overlapping methods) to $+29.3\%$ (fully exclusive). *Spatial coverage is a commodity*; competition’s value there is parsimony and sample-efficiency on noisy real data (it beats the router on traffic at $K=1$ by $+5.1\%$), not dominance.

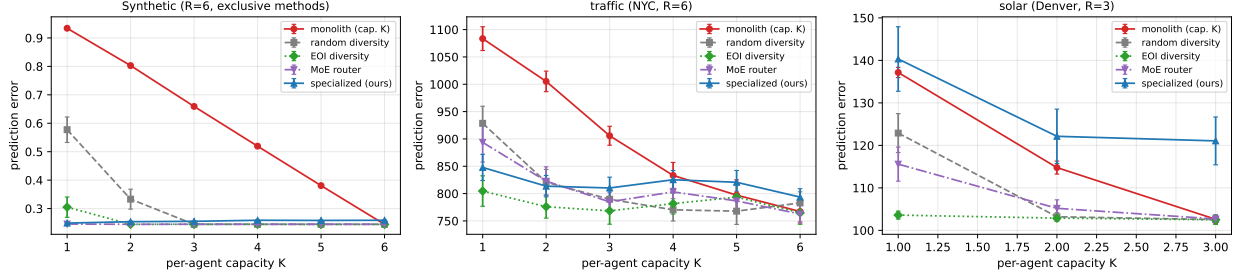


Figure 6 Coverage under bounded capacity: task error versus per-agent capacity K for the five allocation arms (95% CIs), on the synthetic regime-champion stream and real traffic data. The specialized population dominates at tight capacity; purpose-built baselines close the gap as K grows—coverage is a commodity, and competition’s edge is parsimony.

8.4 Results on Retention Under Non-Stationarity

Regimes cycle (a sliding window of $W=3$ of $R=6$ regimes is active per epoch), so every regime goes dormant and reactivates. This is where the five arms dissociate exactly as Principle 1 predicts (Figure 7):

Arm ($K=1$, hard model)	Post-reactivation error	Retains?
Monolith	1.015	no
MoE router	0.928	no
Random diversity	0.603	partially (coverage gaps)
Learned diversity	0.322	yes
GAUSE	0.283	yes

Table 2 Post-reactivation error at $K=1$ under the hard (LRU) capacity model. The two reward-driven arms cluster at the top (forgetting); the three reward-independent arms retain, with GAUSE best.

The population beats the capacity-matched monolith by +33% overall and +71% in the post-reactivation window ($p < 10^{-36}$ at $K=3$), and the router—which matched competition on spatial coverage—fails here decisively (0.928 vs. 0.283 at $K=1$, $p \sim 10^{-35}$; the gap stays significant through $K=4$, $p < 10^{-12}$), closing only as $K \rightarrow R$. The specialized population is essentially flat in K : even $K=1$ agents collectively form a complete persistent memory.

8.5 Ablation: Robustness to the Memory Model

Re-running the entire sweep under the soft interference model (no eviction at all) reproduces the dissociation (Figure 8): the monolith still forgets (post-reactivation error 0.85–0.92 at $K \leq 3$), the specialized population still retains (≈ 0.25 , a +70% gap, $p \sim 10^{-40}$), and the router remains significantly worse than the population (+36% at $K=1$, $p \sim 10^{-8}$; +16% at $K=3$). As the Observation predicts, the router degrades more gracefully under graded interference than under hard eviction—but never closes the gap until $K \rightarrow R$. The separation is a property of reward-driven allocation, not of any particular forgetting mechanism.

8.6 Ablation: The Honest Audit

Three controls scope all of the above. (i) SI is invariant to regime-label shuffling: it measures organization, not structure. (ii) An oracle structure-diagnostic shows four of six domains have a globally dominant method (no exploitable regime structure); on the two structured domains (traffic, solar), causal leakage-free regime labels beat shuffled labels by +15.4% and +12.8%. (iii) With *unlimited* capacity, a single regime-conditioned monolith matches or beats the population: division of labor is a task-accuracy lever *specifically*

Non-stationary regimes (R=6, W=3): population as distributed persistent memory

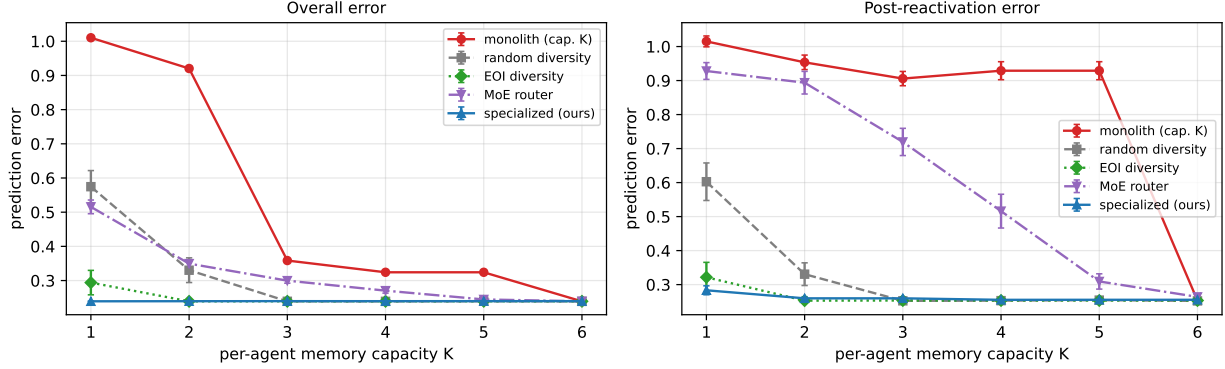


Figure 7 Retention under non-stationarity (hard LRU model, 95% CIs): overall and post-reactivation error versus capacity K for the five arms. The reward-driven arms (monolith, MoE router) pay a persistent forgetting penalty that closes only as $K \rightarrow R$; the reward-independent arms are flat in K .

Robustness: soft interference capacity model (R=6, W=3)

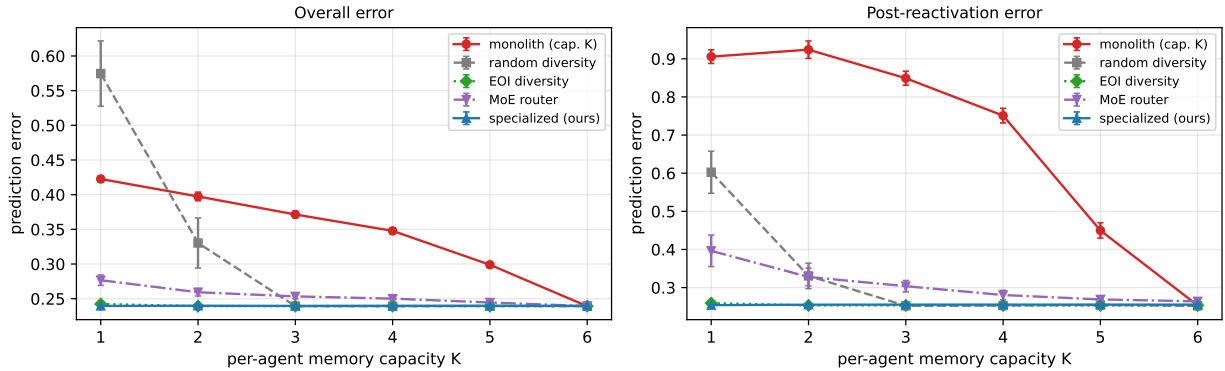


Figure 8 Robustness to the capacity model: the full non-stationary sweep re-run under the soft interference-decay model (no eviction). The reward-independence dissociation survives intact, ruling out an artifact of discrete LRU eviction.

under bounded capacity and non-stationarity, not in general. We consider this audit as much a contribution as the positive results: it states exactly where the mechanism’s value lives.

9 Potential Applications

GAUSE is a mechanism, not a model: it applies wherever (i) the environment decomposes into recurring regimes, (ii) per-learner capacity is bounded, and (iii) regimes go dormant and return. Those three conditions are remarkably common. This section surveys the application domains where we believe the mechanism—and especially its retention guarantee—has the most leverage. Table 3 summarizes the mapping; a deployment recipe follows at the end of the section.

9.1 Elastic Memory for Agentic AI and Mixture-of-Experts Systems

The most direct application is the one our theory engages: **capacity allocation inside AI systems themselves**. Modern agent frameworks accumulate skills, tools, and episodic memories; MoE-based models

Field	Regimes	Learners	Dormancy risk (what is forgotten)
Agentic AI memory	task/skill families	skill modules, expert pools	rarely-invoked skills overwritten by frequent ones
Quantitative finance	market regimes (trend, crisis, range)	strategy pool / capital sleeves	crisis playbooks decay during calm markets
Edge sensing & IoT	seasonal/diurnal patterns, fault modes	on-device models	rare fault signatures lost between occurrences
Robot fleets	zones, task types, failure modes	robots / controllers	seldom-used manipulation skills degrade
Cloud serving	workload classes	autoscaled worker pools	warm capacity for bursty rare workloads
Clinical monitoring	patient subpopulations, rare conditions	model ensemble	rare-condition detectors drowned by common cases

Table 3 Mapping GAUSE onto application domains. In every row the dormancy column is the failure a reward-driven (usage-driven, profit-driven, traffic-driven) allocator suffers and a reward-independent population structurally avoids.

allocate experts to input distributions; and both face the same economics—bounded capacity, a long tail of rarely-activated competences, and usage statistics that concentrate on the recent past. Any eviction or routing policy driven by recency or current utility (LRU caches, usage-weighted distillation, gates trained on task loss) is a reward-chasing allocator in our taxonomy, and Principle 1 predicts exactly its failure: the skill you need *next quarter* is the one producing no usage signal *today*. A GAUSE layer offers a coordination-free alternative: let skill modules or experts *compete* for task families during training or fine-tuning, let identities converge, and then rely on the converged assignment—rather than a still-learning gate—to decide who holds what. This realizes the gate-freezing prescription of MoE continual-learning theory [15] without choosing a freezing schedule, because convergence of the competition *is* the freeze. Concretely: expert pools in continually-pretrained LLMs, skill libraries in agent frameworks, and retrieval indices that must not evict rare-but-critical entries are all candidates.

9.2 Quantitative Finance: Regime-Switching Strategy Pools

Financial markets are the canonical regime-switching environment—trending, mean-reverting, high-volatility, and crisis regimes recur on irregular schedules [10]—and two of our six validation domains (crypto, commodities) are financial. The standard industry practice is reward-chasing by construction: capital and research attention flow to the strategies that performed recently, and strategies that lose relevance are retired. Our results identify the precise cost of that practice. A crisis regime may be dormant for years; a capital allocator driven by recent P&L (the financial analogue of a learned gate) starves the crisis specialists, their infrastructure and calibration decay, and when the regime reactivates the desk pays the full relearning cost—at the worst possible time. A GAUSE-style strategy pool inverts this: strategies compete for regimes, the winner-take-all dynamics assign each strategy a niche, and converged specialists are *retained through dormancy by identity*, not by P&L. The niche bonus λ has a natural reading as a diversification mandate, and the capacity bound K models the real constraint that a single desk cannot stay calibrated to every market state. Three concrete uses: (i) a **regime-specialist ensemble** for return forecasting, where the dormant-regime specialist (e.g., the short-volatility-crash model) is kept warm; (ii) **market-making parameter banks**, where quoting profiles specialized to liquidity regimes are retained across regime cycles; and (iii) **risk-model committees**, where stress-period models must survive long calm stretches—exactly the retention property reward-driven model selection lacks.

9.3 Distributed Sensing and Edge Intelligence

Sensor networks and edge deployments are populations of capacity-limited learners by physical necessity: each device has tight memory and compute, and the environment it models has strong seasonal and episodic

structure (diurnal cycles, weather patterns, rare fault signatures). Centralized coordination is expensive precisely where edge deployment is attractive. GAUSE needs only a local competition signal (which device best predicted the current window), so a fleet can partition pattern-space without a coordinator: some devices become storm specialists, others baseline-drift specialists. The retention property matters most for **rare-event detection**: a fault signature seen once a year is a dormant regime, and a usage-driven model update schedule will overwrite it. Our solar, weather, and air-quality domains are direct evidence that the mechanism organizes correctly on exactly this kind of data.

9.4 Multi-Robot Systems and Task Allocation

Warehouse fleets, agricultural robots, and search-and-rescue teams need division of labor, and the standard solutions are auction protocols or centrally trained role policies—machinery with communication and training costs. Winner-take-all competition over task types offers a degenerate-free alternative: robots that happen to win deliveries in zone A accumulate affinity for zone A, the fleet partitions the floor, and the partition is maintained without any further message passing. The capacity bound is physical (battery, gripper config, local map memory), and the dormancy story is operational: a robot fleet that reallocates all capacity to today’s task mix forgets the skills (or maps, or calibrations) for the seasonal peak, the rare spill response, the annual inventory. A converged niche structure keeps those specialists ready at the cost of some idle time—exactly the trade our non-stationary experiments quantify.

9.5 Cloud Serving and Elastic Infrastructure

Autoscaling systems allocate workers by current traffic—reward-chasing in its purest form—and the known pathology is the **cold-start penalty** on bursty, rare workload classes: by the time the rare workload returns, no warm worker (no JIT-compiled path, no loaded model, no populated cache) remains. Mapping our arms onto serving: pure autoscaling is the monolith; a learned request router over specialized worker pools is the MoE arm; static over-provisioning per class is random diversity (retains but wastes); and a GAUSE-style pool—workers compete for request classes, converged specialists are pinned regardless of current traffic share—achieves retention with coverage. The overlap-sweep result gives the design guidance: the more the workload classes differ in what serves them well (exclusive methods), the larger the payoff from competitive specialization.

9.6 Clinical Monitoring and Rare-Condition Models

Patient-monitoring ensembles face a stark version of the dormancy problem: common presentations dominate the data stream, and models updated on aggregate performance drift toward the majority, degrading exactly on the rare conditions where model value is highest. Treating patient subpopulations or condition classes as regimes, an ensemble trained under competitive exclusion dedicates members to rare presentations and—critically—retains them even when months pass between occurrences. The honest audit transfers too: in deployments where one model genuinely dominates everywhere (no regime structure), the population adds nothing, and our structure diagnostic (Section 8.6) gives a practical test for whether specialization will pay before committing to it.

9.7 When Not to Use It

The audit section is a user manual in miniature, and we restate it as deployment guidance. GAUSE does *not* pay when: (i) capacity is effectively unlimited—a single regime-conditioned model then matches the population with less machinery; (ii) the environment has no regime structure (one method dominates globally)—our oracle diagnostic detects this case in advance; or (iii) regimes never go dormant—without non-stationarity, retention is free and reward-driven routing is a fine choice. The mechanism’s value concentrates precisely where all three conditions bind, which the applications above are selected to satisfy.

Deployment recipe. (1) Define regimes: a discrete, recurring partition of the input stream (volatility bands, workload classes, seasonal modes). (2) Define the quality signal: any per-step scalar comparable across learners. (3) Set capacity K honestly: how many regimes can one learner stay calibrated on? (4) Run winner-take-all competition with the EG affinity update ($\eta \approx 0.6$, $\lambda \in [0.2, 0.4]$ or 0 for the pure mechanism). (5) Read out the converged assignment and *pin it*: the assignment, not the gate, decides who holds what. (6) Re-open competition only when the regime inventory itself changes.

10 Conclusion and Future Work

We presented GAUSE, a minimal framework in which winner-take-all competition over a niche-affinity simplex makes a population of capacity-bounded learners self-organize into one-per-regime specialists—with no communication, shared rewards, gates, or diversity objectives. The framework’s deepest property is that its converged assignment is *reward-independent*: specialists hold niches by identity, idle through dormancy, and thereby retain what reward-driven allocators—including the standard learned MoE router—systematically forget. Across five capacity-allocation mechanisms, retention tracks reward-independence of assignment five-for-five; across six real-world domains and controlled synthetic tasks, competition matches purpose-built diversity and routing baselines on spatial coverage with none of their machinery, and separates from them decisively on retention (+71% post-reactivation, $p \sim 10^{-35}$, robust to the memory model). An honest audit bounds the claim: with slack capacity and stationary regimes, a regime-conditioned monolith suffices; division of labor pays specifically when capacity binds and the world shifts.

Several directions remain promising. First, a function-approximation demonstration: the reward-independence principle predicts the router’s failure is architecture-agnostic, and validating it with gradient-trained experts on a standard continual-learning benchmark would convert the prediction into evidence. Second, hybrid allocators: a router augmented with an explicit memory-preservation or load-balancing term could in principle reserve idle experts for dormant regimes—our results specify *what objective such a router would need*, and testing one would complete the design space. Third, scaling the population: decentralized competition over hundreds of regimes, where the one-per-regime equilibrium must coexist with agent churn and partial observability. Finally, deployment in the application domains of Section 9—trading-strategy pools, sensor networks, and expert pools inside large models—where the parsimony of competition (no gate to train, no schedule to freeze, no diversity coefficient to tune) is most valuable.

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